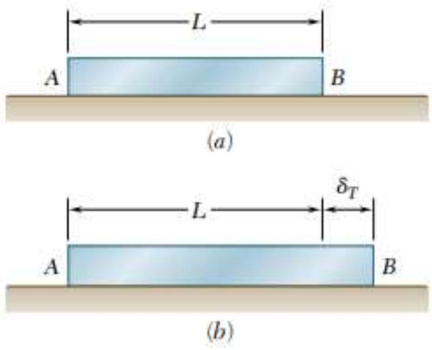
Thermal stresses

Consider a homogeneous rod AB of uniform cross section resting on a smooth horizontal surface. If the temperature of the rod is raised by ΔT, the rod elongates an amount equal to δ_T which is proportional to the length of the bar and the temperature change.

$$\delta_T = \alpha L \Delta T$$

α is known as the coefficient of thermal expansion which is a characteristic of the material



The thermal strain, $\varepsilon_T = \frac{\delta_T}{L} = \alpha \Delta T$

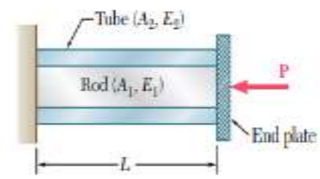
This thermal strain will not introduce any stress in the rod as it is free to expand (or contract). If the expansion(or contraction) is constrained, then only the stresses are developed due to temperature change. Then, they constitute statically indeterminate problems, which are discussed below.

Statically indeterminate problems

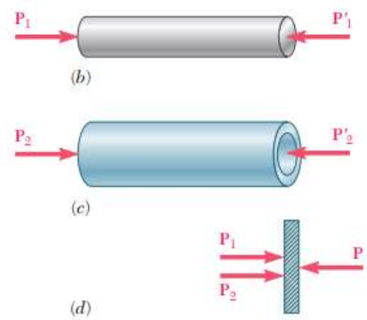
The problems which cannot be solved by the static equilibrium equations alone are classified as statically indeterminate problems. The number of unknowns in such cases will be more than the number of static equilibrium equations. In such situations, we need to use additional equations considering the deformation of the material. These equations are known as compatibility equations.

Example 1

A rod of length $L = 0.75\,m$, cross sectional area $A_1 = 200\,mm^2$ and modulus of elasticity $E_1 = 200\,GPa$ is inserted in a tube of the same length, but of cross sectional area $A_2 = 100\,mm^2$ and modulus of elasticity $E_2 = 120\,GPa$. If the ends are rigidly connected together and a load of 150 kN is applied as shown, determine the deformation of the tube and rod and the compressive stresses on the tube and the rode.



Since, the ends are connected together, the deformation happening to the tube and the rod will be the same, but the load acting on both will be different. Let the force on the rod be P_1 and that on the tube be P_2. Considering the free-body diagram of the cover plate, we write the static equilibrium equation as,



$$P_1 + P_2 - P = 0 \quad \lll\ggg \quad P_1 + P_2 = 150\,kN$$

Since, there are two unknowns, we cannot solve the equation using the static equilibrium equations alone. Hence, this is a statically indeterminate problem. Let us consider the compatibility equation by considering the deformation of the rod and tube. The forces acting on the tube and rod are axial, and hence the deformation of the rode, $\delta_1 = \frac{P_1 L}{A_1 E_1}$ and that of tube $\delta_2 = \frac{P_2 L}{A_2 E_2}$. Since, both deform together, we can write, $\delta_1 = \delta_2$ (compatibility condition)

$$\frac{P_1 L}{A_1 E_1} = \frac{P_2 L}{A_2 E_2} \quad \lll\ggg \quad P_1 = \left(\frac{A_1 E_1}{A_2 E_2}\right) P_2 = \frac{200 \times 10^{-6} \times 200 \times 10^{9}}{100 \times 10^{-6} \times 120 \times 10^{9}} P_2 = 3.33 P_2$$

$$\therefore 3.33 P_2 + P_2 = 150 \times 10^3\,N$$

$$P_2 = 34642\,N \quad \lll\ggg \quad P_1 = 115358\,N$$

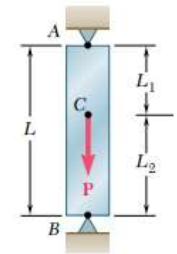
$$\therefore \sigma_1 = \frac{P_1}{A_1} = \frac{115358\,N}{200 \times 10^{-6}\,m^2} = 576.8 \times 10^6 \frac{N}{m^2} \lll\ggg \sigma_2 = \frac{P_2}{A_2} = \frac{34642\,N}{100 \times 10^{-6}\,m^2} = 346.4 \times 10^6 \frac{N}{m^2}$$

The compressive stress in rod= $576.8\,MPa$

The compressive stress in tube= $346.4\,MPa$

Example 2

A bar AB of length $L = 2m$ and uniform cross section of $A = 300\ mm^2$ is kept between two rigid supports at A and B before loading. If a load $P = 100\ kN$ is acting at C which is at $L_1 = 0.75\ m$ from A, determine the stresses in the part AC and BC. The modulus of elasticity of the material is $E = 200\ GPa$.

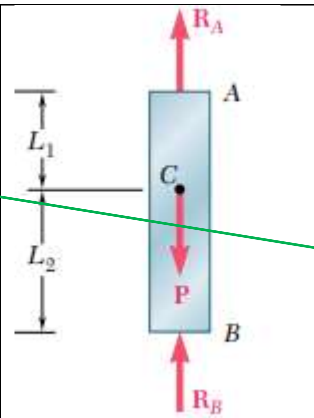


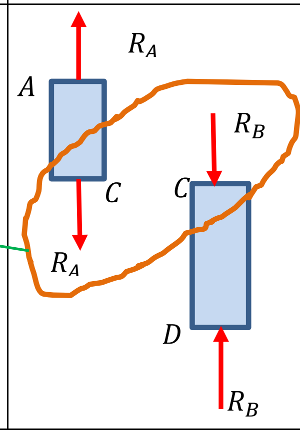
Consider the free-body diagram of the bar.

$$R_A + R_B - 100\ kN = 0$$

$$R_A + R_B = 100 \times 10^3\ N$$

This is a statically indeterminate problem





Let us consider the free body diagram of AC and BC. The force at C on AC should be R_A downwards and that at C on BC should be R_B downwards for their equilibrium. The total force at C (when both parts are joined) will be $R_A + R_B$ which is equal to P. It can be seen from the free-body diagrams that bar AC is in tension and bar BC is in compression.

The elongation of AC,

$$\delta_{AC} = \frac{R_A\ L_1}{AE} = R_A \frac{0.75\ m}{300 \times 10^{-6} m^2 \times 200 \times 10^9\ N/m^2} = 1.25 \times 10^{-8} R_A$$

The compression of BC,

$$\delta_{BC} = -\frac{R_B\ L_2}{AE} = -R_B \frac{1.25\ m}{300 \times 10^{-6} m^2 \times 200 \times 10^9 \frac{N}{m^2}} = -2.08 \times 10^{-8} R_B$$

Since, the bar is kept between two rigid supports, the total change in length of the bar is zero (Compatibility condition)

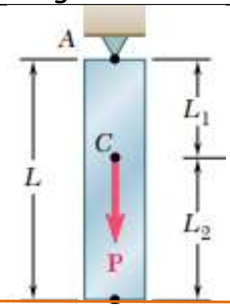
$$\delta_{AC} + \delta_{BC} = 0 \quad \lll\ggg \quad 1.25 \times 10^{-8} R_A - 2.083 \times 10^{-8} R_B = 0 \quad \ggg\ggg \quad R_A = 1.667 R_B$$

$$1.667 R_B + R_B = 100 \times 10^3\ N \ggg\ggg \quad R_B = 37.5 \times 10^3\ N \quad \lll\ggg \quad R_A = 62.5\ \times 10^3\ N$$

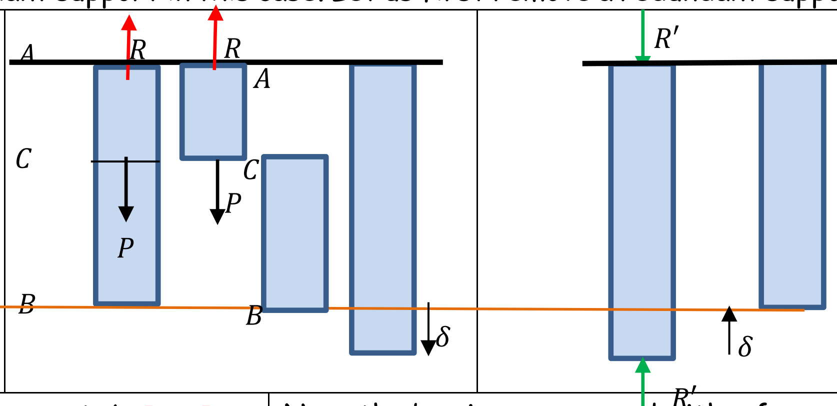
Stress on AC, $\sigma_{AC} = \frac{R_A}{A} = \frac{62.5\ \times 10^3\ N}{300 \times 10^{-6} m^2} = 208.3 \times 10^6\ Pa\ \ (Tensile)$

Stress on BC, $\sigma_{BC} = \frac{R_B}{A} = \frac{37.5\ \times 10^3\ N}{300 \times 10^{-6} m^2} = 125 \times 10^6\ Pa\ \ (compressive)$

We can also solve this problem in another way. It is clear that the problem is statically indeterminate because of the second support. For static equilibrium, only one support is enough. There is a redundant support in this case. Let us first remove a redundant support.



The redundant support is removed



The reaction at the support A, $R = P$

Elongation of the part AC,

$$\delta = \frac{PL_1}{AE} = \frac{100 \times 10^3 N \times 0.75\ m}{300 \times 10^{-6} m^2 \times 200 \times 10^9\ N/m^2}$$

$$= 1.25 \times 10^{-3}\ m$$

$$Stress\ on\ AC = \frac{100 \times 10^3 N}{300 \times 10^{-6} m^2}$$

$$= 333.3 \times 10^6\ Pa\ (Tensile)$$

$$Stress\ on\ BC = 0$$

Now, the bar is compressed with a force R' at the free end so that the bar get compressed by an amount δ sothat the compatibility condition of zero net deformation is satisfied.

$$\delta = \frac{R'L}{AE}$$

$$1.25 \times 10^{-3}\ m = \frac{R' \times 2\ m}{300 \times 10^{-6} \times 200 \times 10^9}$$

$$R' = 37.5 \times 10^3 N$$

$$stress\ on\ bar\ AB = \frac{37.5 \times 10^3 N}{300 \times 10^{-6} m^2}$$

$$= 125 \times 10^6\ Pa\ \ (Compressive)$$

$$Stress\ on\ AC = 125 \times 10^6\ Pa\ \ (Compressive)$$

$$Stress\ on\ BC = 125 \times 10^6\ Pa\ \ (Compressive)$$

Net stress in part AC, $\sigma_{AC} = 333.3 \times 10^6\ Pa - 125 \times 10^6\ Pa\ = 208.3\ MPa\ (T)$

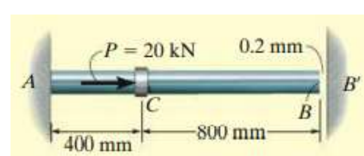
Net stress in part BC, $\sigma_{BC} = 0 - 125 \times 10^6\ Pa\ = -125 \times 10^6\ Pa = 125\ MPa\ (C)$

The reaction at A, $R_A = R - R' = 100 \times 10^3 N - 37.5 \times 10^3 N = 62.5 \times 10^3 N$

The reaction at B, $R_B = 0 + 37.5 \times 10^3 N = 37.5 \times 10^3 N$

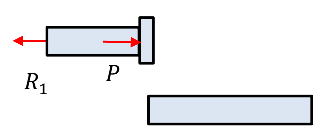
Example 3

The steel rod shown in figure has a diameter of $10\ mm$. It is fixed to the wall at A, and before it is laded, there is a gap of $0.2\ mm$ between the wall at B' and the rod. Determine the reactions at A and B' if the rod is subjected to an axial force of $P = 20\ kN$ as shown. Neglect the size of the collar at C. Take $E = 200\ GPa$



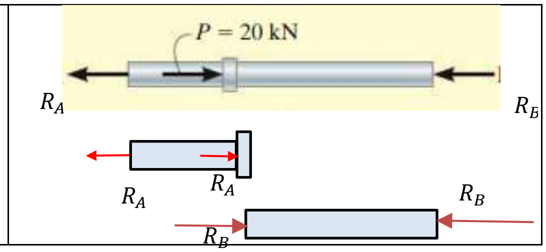
Let us find out the elongation of the bar due to the axial force P. When we draw the free body diagram of AC and CB, we can see that CB is not having any force. From AC, $R_1 = P$

$$\delta = \frac{P\,L_{AC}}{AE} = \frac{20 \times 10^3 N \times 0.4\ m}{\frac{\pi}{4} \times 0.01^2\ m^2 \times 200 \times 10^9 N/m^2} = 0.509 \times 10^{-3} m = 0.509\ mm$$



It is given that there is a gap of $0.2\ mm$ between the rod and the wall. The rod will come and touch the wall only after the elongation exceeds $0.2\ mm$. Then only, there will be reaction at B. Since, the support is rigid, it will not allow the bar to deform further.

Let us consider the free-body diagram of the bar after it touches the support at B.
The equilibrium equation is
$R_A + R_B - P = 0 \lll\ggg R_A + R_B = 20\ kN$
Now, it is a statically indeterminate problem.
Consider the free-body diagram of AC and CB.



Deformation of AC, $\delta_{ac} = \frac{R_A L_{ac}}{AE} = \frac{R_A \times 0.4\ m}{\frac{\pi}{4}\times 0.01^2\ m^2 \times 200\times 10^9 N/m^2} = 2.55\times 10^{-8}\ R_A$

Deformation of CB, $\delta_{bc} = -\frac{R_B L_{cb}}{AE} = -\frac{R_B\times 0.8\ m}{\frac{\pi}{4}\times 0.01^2\ m^2\times 200\times\frac{10^9 N}{m^2}} = -5.10\times 10^{-8}\ R_B$

Once the rod touches the support, the net deformation of the rod $0.2mm$ (compatibility condition).

$$2.55\times 10^{-8}\ R_A - 5.10\times 10^{-8}\ R_B = 0.2\times 10^{-3} \lll\ggg 2.55\ R_A - 5.1R_B = 0.2\times 10^5$$

$$5.1R_A + 2.55R_A = 5.1\times 20\times 10^3 + 0.2\times 10^5 \quad \lll\ggg\ R_A = 15.95\times 10^3\ N \quad \lll\ggg\ R_B = 4.05\times 10^3\ N$$

Before the rod touches the support at B,

$$R_A = 20\times 10^3\ N \quad \lll\ggg\ R_B = 0$$

After it touches the support at B,

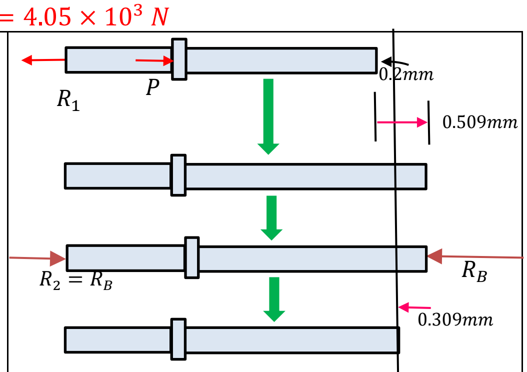
$$R_A = 15.95\times 10^3\ N \quad \lll\ggg\ R_B = 4.05\times 10^3\ N$$

Another way:
If we assume that there is no support at B, the force P will elongate the bar by an amount equal to $0.509\ mm$. Now, the force R_B will compress the bar by amount equal to $0.309\ mm$

$$0.309\times 10^{-3} m = \frac{R_B\ L_{ab}}{AE} = \frac{R_B\times 1.2\ m}{\frac{\pi}{4}\times 0.01^2\ m^2\times 200\times 10^9 N/m^2}$$

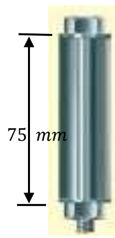
$$R_B = 4.05\times 10^3 N$$

Reaction at A,
$R_A = R_1 - R_2 = 20\times 10^3 - 4.05\times 10^3 N = 15.95\times 10^3 N$



Example 4

A bolt made of $2014-T6$ aluminium alloy ($E_{bolt} = 73\ GPa$) is tightened so that it compresses a cylindrical tube made of $AM\ 1004-T61$ magnesium alloy ($E_{tube} = 45\ GPa$). The tube has a length of $75\ mm$ and an outer radius of $12.5\ mm$, and it is assumed that both the inner radius of the tube and the radius of the bolt are $6.25\ mm$. The washers at the top and bottom of the tube are considered to be rigid and have a negligible thickness. Initially the nut is hand tightened snugly, then using a wrench, the nut is further tightened one-half turn. If the bolt has 40 threads in $50\ mm$ length, determine the stress in the bolt.



The pitch of the bolt thread= axial advancement of bolt when it is turned one complete revolution.

$$p = \frac{50\ mm}{40} = 1.25\ mm$$

The axial movement of the nut when it tightened one-half turn$= 0.625\ mm$

It can be seen that the tube will be in compression and the bolt will be in tension. The equilibrium equation is

$$F_{bolt} - F_{tube} = 0 \quad \lll\ggg \quad F_{bolt} = F_{tube}$$

This is again a statically indeterminate problem. When the nut is tightened, the tube will be compressed and the bolt will be elongated. Therefore, the distance moved by the nut will be the result of compression of tube and elongation of the bolt.

Elongation of the bolt, $\delta_{bolt} = \frac{F_{bolt} \times L}{A_{bolt} \times E_{bolt}} = \frac{F_{bolt} \times 0.075\ m}{\pi \times 0.00625^2\ m^2 \times 73 \times 10^9 N/m^2} = 8.3725 \times 10^{-9}\ F_{bolt}$

Compression of the tube, $\delta_{tube} = \frac{F_{tube} \times L}{A_{tube} \times E_{tube}} = \frac{F_{tube} \times 0.075\ m}{\pi \times (0.0125^2 - 0.00625^2)\ m^2 \times 45 \times 10^9 N/m^2} = 4.5275 \times 10^{-9}\ F_{tube}$

Hence, we can write the compatibility equation as,

$$0.625 \times 10^{-3} m = \delta_{bolt} + \delta_{tube}$$

$$0.625 \times 10^{-3} m = 8.3725 \times 10^{-9}\ F_{bolt} + 4.5275 \times 10^{-9}\ F_{tube}$$

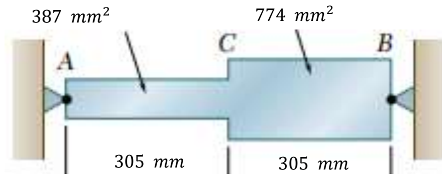
$$F_{bolt} = F_{tube} = 48.5 \times 10^3 N$$

Stress on the bolt, $\sigma_{bolt} = \frac{48.5 \times 10^3 N}{\pi \times 0.00625^2\ m^2} = 395.2 \times 10^6\ N/m^2$ (Tensile)

Stress on the tube, $\sigma_{tube} = \frac{48.5 \times 10^3 N}{\pi \times (0.0125^2 - 0.00625^2)\ m^2} = 131.7 \times 10^6\ N/m^2$(Compressive)

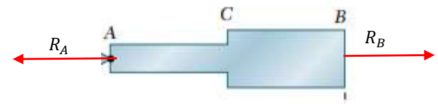
Example 5

Determine the values of the stress in portions AC and CB of the steel bar shown when the temperature of the bar $-50°C$, knowing that a close fit exists at both rigid supports when the temperature is $+75°C$. Use the values $E = 200\ GPa$, $and\ \propto = 6.5 \times 10^{-6}/°C$ for steel.



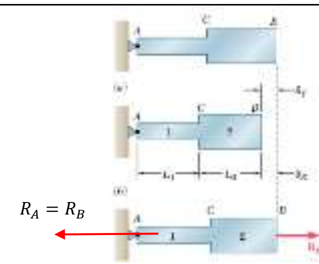
When the temperature drops by $125°C$, the bar contracts. But, the rigid supports will not permit the bar to contract. The free-body diagram of the bar will be as shown. The Equilibrium equation is,

$$R_B - R_A = 0$$



This is a statically indeterminate problem. Hence, let us assume that the support at B is removed and the temperature is decreased by $125°C$. The bar is free to contract now, and the decrease in length of the bar due to the change in temperature

$$\delta_t = \alpha L \Delta T = 6.5 \times \frac{10^{-6}}{°C} \times 0.610m \times 125°C = 0.5 \times 10^{-3}m$$



As the bar is free to contract, there will not be any stress on the bar. Also, please note that the thermal strain does not depend on the cross sectional area.

Now, a force R_B is applied at B, which will pull the bar back to the original position, so that the net deformation is zero. That is, R_B should produce an elongation which is equal to $0.5\ mm$. That is, the compatibility condition is

$$0.5 \times 10^{-3}m = \delta_1 + \delta_2 = \frac{R_B \times L_1}{A_1 \times E_1} + \frac{R_B \times L_2}{A_2 \times E_2} = \frac{R_B \times 0.305m}{387 \times 10^{-6}\ m^2 \times 200 \times 10^9 N/m^2} + \frac{R_B \times 0.305m}{774 \times 10^{-6}\ m^2 \times 200 \times 10^9 N/m^2}$$

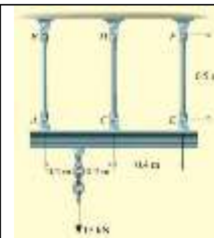
$$R_B = 84.59 \times 10^3\ N$$

The stress on AC, $\sigma_{AC} = \frac{84.59 \times 10^3\ N}{387 \times 10^{-6}\ m^2} = 218.6 \times 10^6\ Pa$

The stress on BC, $\sigma_{BC} = \frac{84.59 \times 10^3\ N}{774 \times 10^{-6}\ m^2} = 109.3 \times 10^6\ Pa$

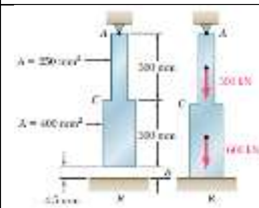
#Exercise 1

The three $A992$ steel bars shown in figure are pin connected to a rigid member. If the applied load on the member is $15\ kN$, determine the force developed in each bar. Bars AB and EF each have a cross sectional area of $50\ mm^2$, and bar CD has a cross-sectional area of $30\ mm^2$.



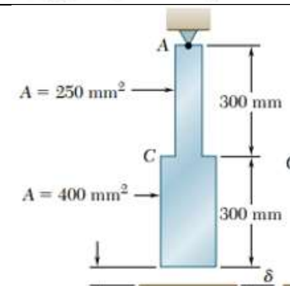
#Exercise 2

Determine the reactions at A and B if a load of $300\ kN$ and $600\ kN$ are acting on the bar as shown.



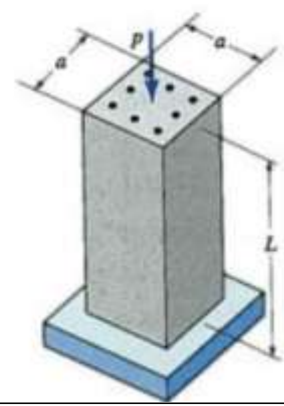
#Exercise 3

The bar shown in the figure is suspended with a gap of $1\ mm$ at the bottom. If the temperature of the bar is raised by $500°C$. Determine the stress developed in the bar.



#Exercise 4

A reinforced concrete column of $2m$ height and $0.5m$ square cross section is constructed with 8 reinforcing steel bars each of diameter $25\ mm$. The column supports a compressive load P applied through a rigid bearing plate. Assuming linear elastic behavior, calculate the maximum permissible value of P if the allowable stresses in steel and concrete are $70\ MPa$ and $8\ MPa$ respectively. Neglect the weight of the pedestal itself, and assume $E_{steel} = 200\ GPa$ and $E_{concrete} = 25\ GPa$



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